

## Rohit Chaki – PES1UG24AM228 – Scenario 17 – Lab 01

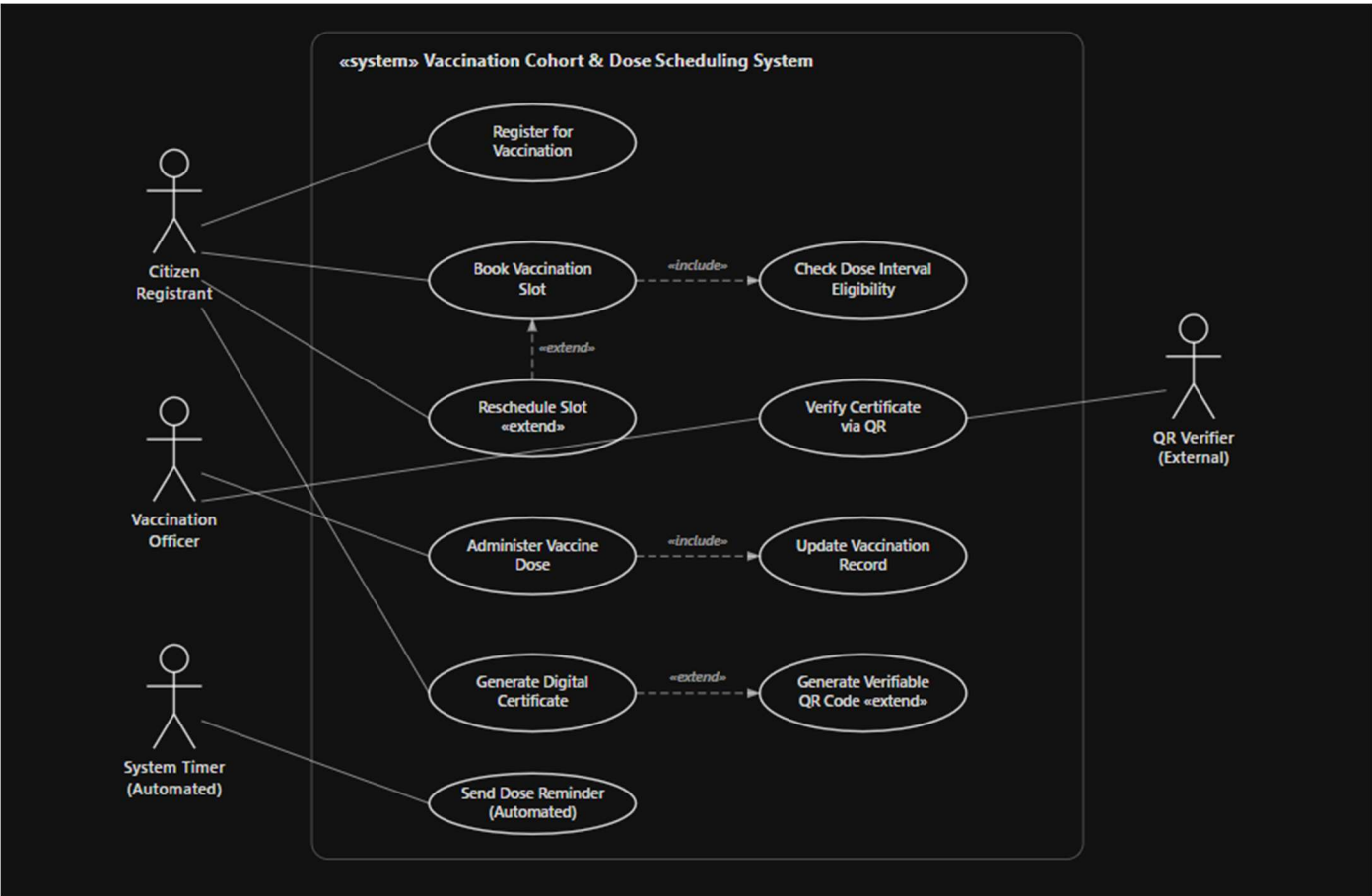
### Functional Requirements:

| ID     | TYPE                      | DESCRIPTION                                                                           | PRIORITY | ACCEPTANCE CRITERIA                                                                                                              | RATIONALE                                                                                                               |
|--------|---------------------------|---------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| FR-001 | Business Rule             | Enforce minimum dose interval (e.g., 28 days) before unlocking next-dose booking.     | High     | Pass: Booking blocked if interval < 28 days with days-remaining shown. Fail: Early slot is bookable.                             | Protects immunization efficacy and adheres to clinical schedules; prevents reduced effectiveness from premature dosing. |
| FR-002 | Process / Workflow        | Allow citizens to search and book real-time slots at vaccination centers.             | High     | Pass: Only open slots for eligible doses are shown; confirmation ID issued. Fail: Overbooking or ineligible dose booking occurs. | Eliminates physical crowding; real-time inventory prevents center overload and wasted citizen trips.                    |
| FR-003 | Security / Data Integrity | Generate a digitally signed certificate with a verifiable QR code per completed dose. | High     | Pass: Certificate contains dose metadata and a QR signed by the health authority. Fail: Unsigned or tampered QR payload.         | Prevents certificate forgery for travel and workplace mandates; ensures cross-border trust via cryptographic proof.     |
| FR-004 | Data / Classification     | Auto-assign registrants to demographic cohorts for phased rollout access control.     | Medium   | Pass: Cohort assignment matches profile; phase gates block ineligible bookings. Fail: Misassignment grants early access.         | Prioritizes high-risk groups during supply shortages; enables equitable, epidemiologically sound distribution.          |
| FR-005 | Notification              | Send automated reminders (SMS/app) before the next dose eligibility window.           | Medium   | Pass: Reminder sent 48h before eligibility with booking link via preferred channel. Fail: Missing or late reminder.              | Reduces Dose-1-to-Dose-2 dropout; cuts manual follow-up burden and keeps citizens within optimal immunization windows.  |

### Non-Functional Requirements:

| ID      | TYPE                       | DESCRIPTION                                                                    | PRIORITY | ACCEPTANCE CRITERIA                                                                                                         | RATIONALE                                                                                                              |
|---------|----------------------------|--------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| NFR-001 | Performance & Security     | Verify digitally signed QR codes in < 150 ms, supporting offline/online modes. | High     | Pass: Latency and security benchmarks pass at 10,000 concurrent verifications. Fail: > 150 ms or missed invalid signatures. | Enables high-throughput verification at airports and venues; offline mode supports rural connectivity gaps.            |
| NFR-002 | Availability & Scalability | Maintain 99.9% uptime and auto-scale to 100,000 concurrent users within 60 s.  | High     | Pass: Zero data loss during failover; no degradation under peak load. Fail: > 0.1% monthly downtime or booking failures.    | Handles announcement-driven rushes (e.g., eligibility expansion); prevents public frustration and immunization delays. |

UML Diagram:



Use-Case Flow Specification

1. Use-Case Header

| Attribute     | Detail                                                                                                                                                                                                                                 |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use-Case ID   | UC-002                                                                                                                                                                                                                                 |
| Use-Case Name | Book Vaccination Slot                                                                                                                                                                                                                  |
| Actor(s)      | Citizen Registrant (Primary)                                                                                                                                                                                                           |
| Description   | The citizen searches for available vaccination centers and time slots for their eligible dose, selects a preferred option, and confirms the booking. The system enforces dose interval rules and cohort eligibility before confirming. |
| Trigger       | Citizen clicks "Book Slot" from their dashboard or receives a dose-reminder notification.                                                                                                                                              |

2. Preconditions

1. The citizen has completed registration (UC-001) and holds a valid registrant profile in the system.
2. The citizen's demographic cohort has been activated for the current rollout phase (FR-004).
3. The citizen has an active, unexpired session (authenticated).
4. At least one vaccination center in the citizen's registered geographic zone has published open slots.

### 3. Postconditions

| Outcome            | Postcondition                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Success            | A confirmed booking record is created; slot inventory at the selected center is decremented by 1; citizen receives a confirmation with QR check-in code and center details. |
| Failure (Interval) | No booking created; citizen is shown days remaining until eligibility with an option to "Notify Me When Eligible."                                                          |
| Failure (No Slots) | No booking created; citizen is shown nearest alternative centers/dates with a waitlist option.                                                                              |

### 4. Main Success Scenario

| Step | Actor Action                                                             | System Response                                                                                                                                                         |
|------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Citizen selects "Book Vaccination Slot" from the dashboard.              | System displays the dose-selection screen (Dose 1 / Dose 2 / Booster) based on the citizen's vaccination history.                                                       |
| 2    | Citizen selects the dose they wish to book.                              | System invokes «include» <b>Check Dose Interval Eligibility</b> (FR-001). System validates the citizen's cohort phase (FR-004).                                         |
| 3    | —                                                                        | System queries real-time slot inventory for centers within the citizen's registered zone and displays a searchable list with date, time, center name, and vaccine type. |
| 4    | Citizen filters by preferred date range and center, then selects a slot. | System locks the selected slot temporarily (5-minute hold) and presents a booking summary.                                                                              |
| 5    | Citizen reviews details and confirms the booking.                        | System commits the booking, decrements center inventory, generates a unique booking reference ID, and creates a calendar-entry QR check-in code.                        |
| 6    | —                                                                        | System sends confirmation via SMS and in-app notification containing center address, time, and check-in QR.                                                             |
| 7    | Use case ends successfully.                                              | —                                                                                                                                                                       |

### 5. Alternate Flow A — Reschedule Request (extends UC-002)

| Step | Condition                                                                                     | Actor Action                                                          | System Response                                                                                                                                          |
|------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| A.1  | Citizen already holds a confirmed future booking and selects "Reschedule" from the dashboard. | Citizen selects "Reschedule" and is prompted for a reason (optional). | System cancels the existing booking, restores the slot inventory at the original center, and returns the citizen to Step 3 of the Main Success Scenario. |

| Step | Condition                                         | Actor Action                              | System Response                                                                                                                        |
|------|---------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| A.2  | —                                                 | Citizen proceeds with new slot selection. | System applies the same eligibility and inventory checks as in the main flow.                                                          |
| A.3  | Reschedule limit exceeded (e.g., >2 reschedules). | —                                         | System displays a warning: "Reschedule limit reached. Please contact support or proceed to the originally booked slot." Use case ends. |

## 6. Exception Flow E1 — Dose Interval Violation

| Step | Condition                                                                                                           | System Response                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| E1.1 | At Step 2, the citizen selects Dose 2 (or Booster) but the minimum interval from the previous dose has not elapsed. | System blocks booking and displays: "You are not yet eligible for Dose 2. You may book after [eligible date, 28 days from Dose 1]." |
| E1.2 | —                                                                                                                   | System offers a "Remind Me" button; if clicked, a reminder task is scheduled (FR-005). Use case ends.                               |

## 7. Exception Flow E2 — No Available Slots

| Step | Condition                                                                               | System Response                                                                                                               |
|------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| E2.1 | At Step 3, no centers in the citizen's zone have open slots for the eligible dose type. | System displays: "No slots available in your zone. Join the waitlist or view nearby zones?"                                   |
| E2.2 | Citizen selects "Join Waitlist."                                                        | System adds citizen to prioritized waitlist (by cohort and registration date) and confirms waitlist placement. Use case ends. |

## 8. Relationship Mapping (for Diagram Traceability)

- **«include» Check Dose Interval Eligibility:** Mandatory validation invoked during Step 2 of every booking attempt (FR-001).
- **«extend» Reschedule Slot:** Optional extension triggered only when a citizen already holds a booking and explicitly requests a change (Alternate Flow A).